DIGITAL ELECTRONICS LAB (ESC392)

Lab Assignment 7 — Design a Mini Automation System

Automatic Parking Barrier Controller

Submitted by: Ritam Maity
Department: CSE
Institute: Institute of Engineering & Management, Kolkata

1. GenAI Model Name

ChatGPT (GPT-6)

2. Prompt

"Design a simple automatic parking barrier controller using digital electronics. Define the inputs and outputs, explain the operating conditions, provide the finite state machine (FSM), state table and logic implementation, and describe how to simulate and verify the circuit in Logisim Evolution."

3. Response

Aim

To design and simulate an automatic parking barrier controller using combinational and sequential logic in Logisim Evolution.

System Description

The system automatically opens a parking barrier when a vehicle is detected and closes it after the vehicle passes. A clock controls the sequence of operations.

Inputs

- **V:** Vehicle detected (1 = vehicle present)
- **C:** Clock signal
- **R:** Reset signal

Outputs

- **B:** Barrier status (1 = open, 0 = closed)

Operating States

- **S0 — CLOSED:** The barrier remains closed until a vehicle is detected.
- **S1 — OPEN:** The barrier opens when a vehicle is detected and remains open while the vehicle is present.

- **S2 — WAIT:** After the vehicle leaves, the system waits for the next clock transition before closing the barrier.

State Table

Present State	Vehicle V	Next State	Barrier B
CLOSED	0	CLOSED	0
CLOSED	1	OPEN	1
OPEN	1	OPEN	1
OPEN	0	WAIT	1
WAIT	1	OPEN	1
WAIT	0	CLOSED	0

Expected Behaviour

1. Initially, the barrier is closed.
2. When a vehicle is detected, the barrier opens.
3. While the vehicle is present, the barrier stays open.
4. After the vehicle leaves, the controller waits one clock cycle and then closes the barrier.
5. The reset input returns the controller to the CLOSED state.

4. Verification by Logisim Evolution

The circuit should be tested using the following cases:

Test	Vehicle V	Expected Barrier
Initial/reset state	0	Closed (0)
Vehicle arrives	1	Open (1)
Vehicle remains	1	Open (1)
Vehicle leaves	0	Remains open during WAIT
Next clock in WAIT	0	Closed (0)

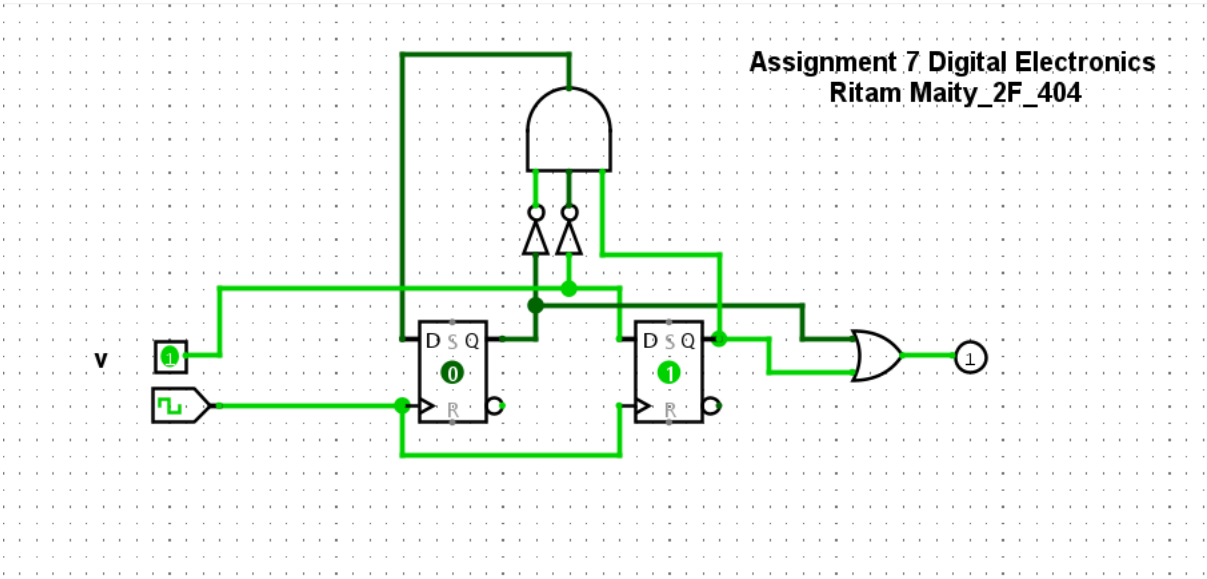
The simulation is considered successful when the observed outputs match the expected results for all test cases.

5. Circuit:

The circuit consists of:

- Two D flip-flops to store the three FSM states.
- Combinational logic gates to calculate the next state.
- A clock input to advance the state.
- A reset input to initialize the system.
- An output indicator LED representing the barrier.

Circuit implementation: Use Logisim Evolution to build the FSM from the state table. Connect the D inputs according to the derived next-state equations, and connect the output logic to the barrier LED.



6. Output

The expected output is:

- Barrier LED OFF when the state is CLOSED.
- Barrier LED ON when the state is OPEN or WAIT.
- Barrier LED OFF again after the vehicle leaves and the controller completes the WAIT state.

7. Discussion

The automatic parking barrier controller demonstrates the practical application of combinational and sequential logic. The FSM stores the current operating state, while the input signal and clock determine the next state. The simulation helps verify the expected behaviour of the barrier under different vehicle-detection conditions.

8. GitHub Link

GitHub Repository: https://github.com/Ritam-Maity/Digital-Electronics-Lab